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## Milestone 1 - Data Description

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### Abstract

This project analyzes the CDC Diabetes Health Indicators dataset derived from the BRFSS to examine associations between lifestyle, access to care, and diabetes. The dataset contains 23 columns including one identifier, 15 binary indicators, four ordered categorical fields, and three numerical measures. The cleaning process was minimal because there are no missing values in the dataset to begin with,

### 1. Data Description

The CDC Diabetes Health Indicators dataset compiles Behavioral Risk Factor Surveillance System (BRFSS) responses from U.S. adults into a large, structured table designed for risk modeling. It contains about a quarter million records with 20+ health-related variables that are mostly binary or ordinal, covering cardiometabolic conditions (such as high blood pressure and high cholesterol), health behaviors (such as smoking and physical activity), self-reported status (including general, mental, and physical health), functional limits (such as difficulty walking), and basic demographics such as age group, sex, education, and income. Body mass index shows up as a numeric field, while most of the other fields use 0/1 or small integer codes that reflect the survey answers. The label indicates whether a person has diabetes based on the answers of their survey, and the other columns combine medical risks (like blood pressure and BMI) with care access and lifestyle factors (like checkups, smoking, and exercise) to reflect everyday influences on diabetes. In general, the dataset presents a glimpse of population health indicators that are reliable for classification and exploratory epidemiologic work.

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*Proceedings of the 42<sup>nd</sup> International Conference on Machine Learning*, Vancouver, Canada. PMLR 267, 2025. Copyright 2025 by the author(s).

### 2. Variable Descriptions

The dataset has 23 variables total: 1 identifier (ID), 15 binary indicators (Diabetes\_binary, HighBP, HighChol, CholCheck, Smoker, Stroke, HeartDiseaseorAttack, PhysActivity, Fruits, Veggies, HvyAlcoholConsump, AnyHealthcare, NoDocbcCost, DiffWalk, Sex), 4 ordered categorical (“factor”) fields (GenHlth 1–5, Age 13 buckets from 18–24 to 80+, Education 1–6 from no school to college 4+ years, Income 1–8 from <\$10k to ≥\$75k), and 3 numerical counts/continuous fields (BMI, MentHlth, PhysHlth). For modeling, the working feature set is 22 columns (everything except ID): 15 binary, 4 ordered categorical, and 3 numeric. For context, Diabetes\_binary shows whether the respondent reports diabetes (prediabetes/diabetes coded as 1), serving as the prediction target. ID is just a unique row key. HighBP and HighChol show lifetime diagnoses of high blood pressure and high cholesterol, while CholCheck records whether cholesterol was checked within the past five years (basic cardiometabolic screening behavior). BMI is body mass index from height/weight and acts as a proxy for adiposity. Smoker represents ever smoking at least 100 cigarettes (standard BRFSS “ever smoker”), a risk factor for vascular disease that tracks with diabetes risk. Stroke and HeartDiseaseorAttack (CHD or MI) represents major cardiovascular history that co-travels with metabolic disease. PhysActivity shows any leisure time physical activity in the last 30 days, and the diet markers Fruits and Veggies represent whether fruit and vegetables are consumed at least once per day; simple indicators for healthier eating. HvyAlcoholConsump records heavy drinking (men >14, women >7 drinks/week), which is relevant to metabolic and liver health. AnyHealthcare shows whether the person has any health coverage, while NoDocbcCost records whether cost prevented a needed doctor visit in the past year (both provide cognizance of access and potential delays in diagnosis or management). GenHlth is a five-level self-rating of overall health (excellent→poor). MentHlth and PhysHlth count days (0–30) in the past month when mental or physical health was “not good,” which often correlate with functional status and chronic disease burden. DiffWalk records serious difficulty walking or climbing stairs, a limitation linked to frailty and advanced disease. Demographics—Sex (0=female, 1=male), Age (13 ordered groups), Education (1–6 ladder of attainment), and Income (1–8 income brackets) all

provide socioeconomic and life-stage context; these are all useful but also sensitive attributes that should be handled carefully in analysis and reporting.

### 3. Challenges for Analysis

The first step in preparing the dataset for analysis was to check for any null or missing values. Using `df.isnull().sum()`, we confirmed that there were no missing entries in any of the columns. All the columns returned 0 as the sum for null values. There was a possibility that the data could have had empty strings and single spaces but there were none in this data set. The lambda function to check for the empty strings and the `isnull()` helped to ensure that this data set was not missing any values. It was also important to verify that each column contained more than one unique value, to make sure every variable contributed meaningful information. This was confirmed by examining the number of unique values in each column using `df.nunique()`, which showed that all columns had multiple unique values. To get an initial overview of the dataset, we printed the shape of the DataFrame as well as the first and last few rows using `df.head()` and `df.tail()`.

Next, we reviewed the data types of the columns to improve the dataset's usability. When the data was initially loaded without pandas, many columns were either binary or integer types. However, after converting the dataset into a data frame using pandas, all columns were seen as integers by pandas. To make the dataset easier to understand, we converted the columns representing binary features from numeric codes (0 and 1) to more descriptive string values. Specifically, binary columns such as `HighBP`, `HighChol`, `CholCheck`, `Smoker`, `Stroke`, `HeartDiseaseorAttack`, `PhysActivity`, `Fruits`, `Veggies`, `HvyAlcoholConsump`, `AnyHealthcare`, `NoDocbcCost`, and `DiffWalk` were mapped so that 0 became 'No' and 1 became 'Yes'. This change helped make the data more intuitive to interpret.

Additionally, some categorical variables that used numeric codes were updated with descriptive labels to improve readability. For example, the `Sex` column was mapped from 0 and 1 to the strings 'Female' and 'Male', respectively, making the demographic data clearer. The `GenHlth` column, which rates general health on a scale from 1 to 5, was converted into categories such as 'Excellent', 'Very Good', 'Good', 'Fair', and 'Poor'. The `Age` variable was also mapped from numeric codes to meaningful age ranges: 1 to '18-24', 2 to '25-29', 3 to '30-34', 4 to '35-39', 5 to '40-44', 6 to '45-49', 7 to '50-54', 8 to '55-59', 9 to '60-64', 10 to '65-69', 11 to '70-74', 12 to '75-79', and 13 to '80+'. This mapping provided a clear and easy-

to-understand representation of age groups. Finally, the education and income columns were converted from numeric codes to meaningful categories that reflect education levels (from no schooling to college graduate) and income brackets (from under \$10,000 to \$75,000 or more). These mappings improved the overall readability of the dataset, making it easier to analyze and interpret.

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## Milestone 2 - Pre-Analysis Plan

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### 4. Method Overview

We will train a small, defensible set of models: multinomial logistic regression with L2 as our linear reference, k-Nearest Neighbors to test local structure, and a decision tree with depth and leaf controls for simple nonlinear splits. We will keep binary flags as binary, respect true ordering for ordinal fields, one-hot multi-category inputs for logistic regression, and standardize continuous variables (BMI, mental-health days, physical-health days, sleep, etc ) inside pipelines. PCA will be fitted only inside cross-validation folds and only for kNN to stabilize distances without leakage. We will use a stratified 80/20 train-test split, select models with 5-fold stratified cross-validation on the train split with fixed seeds, and put all preprocessing in sklearn Pipelines so transforms are learned only from training folds. We will save the fold indices for exact reruns. Our searches target the capacity knobs: C on a log scale from  $10^{-3}$  to  $10^3$  for logistic regression, odd k from 3 to 51 and weight scheme for kNN, and max depth {3, 5, 10, 15, 20}, minimum leaf {1, 5, 10}, and criterion such as gini, entropy for trees. We will choose the smallest k or simplest depth within one-standard-error of the best cross-validated score to reduce variance and bias. We will also generate a learning curve and a calibration curve for the selected model to check data adequacy and probability quality as part of the training and validation diagnostics.

### 5. Models and Justification

We start with multinomial logistic regression with L2. It gives a clear linear reference and stable training on large tabular data. L2 regularization controls variance and helps when inputs correlate (for example, MentHlth and PhysHlth often move together). We will fit the true multinomial (softmax) rather than one-vs-rest so the three classes share one joint model, and we will use `class_weight='balanced'` if class counts are uneven. This model also produces usable probabilities, so

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we will check calibration with a reliability curve and the Brier score to support any risk statements we make.

We add k-Nearest Neighbors to test for local structure. kNN can capture simple nonlinear boundaries without a parametric form, but it needs scaled features or variables like BMI will dominate distance. We will standardize continuous features, compare uniform and distance weighting, and sweep odd k values from small to large to expose the bias-variance trade-off. Because high-dimensional distances can be noisy, we will try a light PCA front-end with a small component range inside cross-validation only for the kNN pipeline. If nearby points in feature space share diabetes status, kNN should beat or match logistic regression on macro-F1; if not, the k curve will plateau and tell us to keep the model simple.

We include a CART decision tree to capture axis-aligned splits and low-order interactions. Trees handle mixed binary and ordinal inputs, give readable rules, and can surface interactions that linear models miss. The main risk is overfitting, so we will control depth and minimum leaf size, compare gini and entropy, and watch the gap between train and cross-validation scores. We may enable `class_weight='balanced'` for the tree if minority classes suffer recall. We will also report permutation importance for the selected tree so any “what matters” claim is grounded and not biased by raw impurity scores.

### 6. Feature Engineering

We will keep binary flags as binary and preserve their meaning (for example, Smoker, PhysActivity). For multi-level responses like general health, education, and income bands, we will respect ordering when it is real and document the choice in the appendix. In the logistic pipeline, we will one-hot multi-category variables to avoid imposing a fake linear distance between categories, and if an ordinal relationship is well supported we will also test an ordinal code and compare results. We will standardize continuous variables that drive distances (BMI, MentHlth, PhysHlth, SleepTime) inside the pipeline so scaling is learned only from training folds. Trees will use the raw encodings because they are scale-invariant. PCA will be used only in the kNN branch, fitted inside each fold, and the number of components will be chosen by cross-validated macro-F1 since our goal is classification accuracy, not reconstruction.

## 7. Training, Validation, and Search

We will use a stratified 80/20 train–test split to preserve class ratios. We will select models with 5-fold stratified K-Fold on the training set and fix a random seed so folds are reproducible. All preprocessing (scalers, encoders, PCA) will live in sklearn Pipelines attached to the estimator, so each fold fits transforms on its own training slice and never sees the validation slice. We will save the fold indices and the seed to the repo for exact reruns. Our searches target the knobs that control capacity. For logistic regression, we will sweep C on a log scale from  $10^{-3}$  to  $10^3$  with L2 and a multinomial solver. For kNN, we will sweep odd k from 3 to 51 and toggle uniform versus distance weighting; for the tree we will sweep max depth (3, 5, 10, 15, 20), minimum leaf (1, 5, 10), and criterion (gini, entropy). We will pick the smallest k or simplest depth within one standard error of the best cross-validated macro-F1 to reduce variance, then confirm with a learning-curve check and a calibration curve, so our final probabilities and error bars are justified, not assumed.

## 8. Evaluation and Decision Rule

Our objective is to develop a compact, well-calibrated classifier that balances overall predictive performance with model simplicity and reliable probability estimates suitable for downstream risk assessments.

We use macro-F1 on stratified 5-fold cross-validation on the training set as our primary selection metric. We will apply the one-standard-error rule, selecting the simplest model (lowest capacity: smallest k for kNN or shallowest depth for trees) whose mean CV macro-F1 is within one standard error of the best-performing model.

To break ties or veto models, we consider calibration, parsimony, stability, and fairness. We prefer models with lower Brier scores and visually acceptable reliability curves. If models have indistinguishable macro-F1, we select the better-calibrated model. Among similarly performing models, we favor logistic regression (interpretable coefficients) or smaller decision trees. We also prefer models with lower CV standard deviation for stability. Models that underperform systematically on protected subgroups (age bands, sex, income, or education) are deprioritized unless fairness mitigation strategies succeed.

We determine classification thresholds and perform probability post-processing by evaluating ROC/PR curves and F1 trade-offs on a held-out validation slice or nested cross-validation. We use 0.5 as the default threshold, but may adjust it to maximize balanced accuracy or macro-F1 for specific applications. If probabilities are poorly calibrated, we apply Platt scaling or isotonic regression via CalibratedClassifierCV, reporting both raw and calibrated probabilities.

ties.

For final evaluation, we assess the selected model once on the held-out 20% test set, reporting macro-F1, per-class F1, balanced accuracy, precision, recall, one-vs-rest ROC AUC, PR AUC, and Brier score. We compute 95% bootstrap confidence intervals (2000 stratified bootstrap samples) for key metrics. When appropriate, we use paired statistical tests for model comparisons (DeLong for AUC, McNemar for paired classification errors, or paired bootstrap for other metrics). We emphasize effect size and uncertainty rather than p-values alone.

The final model is supposed to satisfy the one-standard-error rule on CV macro-F1, meet secondary criteria (calibration and subgroup parity), and demonstrate acceptable uncertainty (narrow confidence intervals) on the test set. If no model meets parity or calibration requirements, we report the trade-offs and recommend mitigation strategies such as reweighting, constrained optimization, or removing sensitive attributes.

## 9. Implementation Details

We will split the data into an 80/20 stratified train/test set and fix all randomness with a constant seed for reproducibility, perform 5-fold stratified cross-validation on the training set using a fixed random state, and save fold indices and seeds in the repository to ensure exact reruns are possible.

We build preprocessing pipelines using scikit-learn’s Pipeline and ColumnTransformer. For numeric features such as BMI, mental health days, physical health days, and sleep time, we first impute missing values using the median, then standardize the features for logistic regression and kNN (trees do not require scaling). We are going to keep binary flags as binary and preserve their meaning. Ordinal features such as general health, age, education, and income preserve their numeric order for trees and logistic regression. For logistic regression, we will one-hot encode categorical variables. The categorical pipeline for logistic regression uses an encoder that ignores unknown categories. For kNN, we may include PCA fitted inside cross-validation, with the number of components tuned during the search. All transformations are going to live inside pipelines to avoid data leakage.

For model training and hyperparameter tuning, we use logistic regression with multinomial loss and L2 regularization, sweeping C values on a logarithmic scale. We apply balanced class weights if necessary. For kNN, we sweep odd values of k from 3 to 51 and test both uniform and distance weighting, optionally tuning PCA components within cross-validation. For decision trees, we sweep max depth (3, 5, 10, 15, 20), minimum samples per leaf (1, 5, 10), and splitting criterion (Gini), and we enable balanced class weights

when needed. We wrap all estimators in GridSearchCV or randomized search with all preprocessing included, using all available cores and fixed seeds.

When calibration is required, we apply Platt scaling or isotonic regression using CalibratedClassifierCV with cross-validation. We select thresholds based on maximizing validation macro-F1 or another contextually relevant metric.

For interpretability, we report permutation feature importance for the final tree using 100 repeats. For logistic regression, we present coefficients as odds ratios with bootstrap confidence intervals.

## 10. Risks, Ethics, and Limitations

There are key risks and limitations present in our analysis. The BRFSS dataset is self-reported and cross-sectional, which introduces potential biases such as recall error, social desirability bias, and measurement error, for example in self-reported height and weight. While the survey provides sampling weights, ignoring them can lead to biased population-level inferences. We are going to take these limitations into account when interpreting the results.

Our work is strictly predictive and associative. We will avoid causal language, such as stating that one factor causes diabetes, unless supported by appropriate causal methods or longitudinal data. We also are not going to publish the ID column or any row-level data. To protect participant privacy, we will aggregate small subgroups in stratified reports and avoid displaying cell counts that could lead to re-identification.

Age, sex, education, and income are all sensitive features. While including them may improve predictive performance, it can also introduce risks of unfair or discriminatory outcomes. We report subgroup metrics, such as true positive rate, false positive rate, and calibration, and consider fairness interventions when necessary, including reweighting, constrained optimization, or removing sensitive attributes for deployment decisions. The cross-sectional, self-reported nature of the data limits temporal generalizability, and our models may be vulnerable to distribution shifts. Rare subgroups can produce unstable estimates; we address this by computing bootstrap confidence intervals and avoiding over-interpretation of small-n differences.

## 11. Planned Tables and Figures

### Tables:

1. Dataset summary: variable name, type (binary/ordinal/numeric), levels, mean/SD or proportion, missingness.
2. Class balance and sample sizes: counts and percent-

ages by diabetes class and key demographics (age bands, sex).

3. CV model selection results: mean  $\pm$  std CV macro-F1 for each model or hyperparameter group; mark chosen model via one-standard-error rule.
4. Test performance: final metrics on test set with 95% bootstrap CIs (macro-F1, per-class F1, balanced accuracy, ROC AUC, PR AUC, Brier).
5. Subgroup performance: sensitivity, specificity, and calibration metrics by age, sex, income, education.
6. Feature importance or coefficients: logistic odds ratios (with CIs) and permutation importances for the tree.

### Figures:

1. Feature distributions: histograms or density plots for numeric features (BMI, MentHlth, PhysHlth) and bar plots for key binaries.
2. Correlation heatmap: pairwise correlations to highlight collinearity.
3. Confusion matrix: normalized confusion matrix on the test set.
4. ROC and PR curves: one-vs-rest ROC/PR for selected models.
5. Calibration plot: reliability curve with Brier score annotation.
6. Learning curve: train vs validation macro-F1 over increasing training set sizes.
7. Feature importance or coefficients: side-by-side importance from tree and logistic (odds ratios).
8. SHAP summary plot: if SHAP is used, show top features and directionality.
9. Subgroup performance visualization: small multiples showing sensitivity/specificity or ROC across groups, annotated with sample sizes.
10. Appendix figure: partial dependence plots for top continuous predictors (e.g., BMI, MentHlth).

Each table or figure will state whether metrics are weighted or unweighted, whether probabilities were calibrated, and the exact sample used (train, CV, or test). Subgroup figures will include sample size annotations.

# Milestone 3 - Initial Results

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## 12. Model Implementation and Training Details

We implemented logistic regression (multinomial with L2 regularization), k-Nearest Neighbors (kNN), and CART decision tree models as outlined in the pre-analysis plan. Preprocessing pipelines included:

- Standardization of numeric features (BMI, MentHlth, PhysHlth)
- One-hot encoding for categorical variables in logistic regression
- Binary flags preserved as-is
- PCA applied only for kNN within cross-validation folds

All models were trained using 5-fold stratified cross-validation on the training set (80% of the dataset), with fixed random seeds to ensure reproducibility. Hyperparameter grids were searched using GridSearchCV, and the final model was selected using the one-standard-error rule on macro-F1.

## 13. Evaluation Benchmark

### 13.1. Dataset

The dataset is the CDC Diabetes Health Indicators dataset derived from BRFSS, containing 23 variables and approximately 250,000 records. The working feature set includes 22 predictors (excluding ID), combining binary, ordinal, and numeric variables.

### 13.2. Performance Metrics

Evaluation metrics include:

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- Macro-F1 (primary selection metric)
- Balanced accuracy
- One-vs-rest ROC AUC
- PR AUC
- Brier score
- Per-class F1

Metrics are computed on the held-out 20% test set, with 95% bootstrap confidence intervals (2000 stratified resamples). Calibration curves and reliability plots are used to evaluate probability quality. Subgroup metrics (age, sex, income, education) are also calculated to assess fairness.

### 13.3. Cross-Validation Summary

| Model               | Best Parameters                                                                  | CV F1  |
|---------------------|----------------------------------------------------------------------------------|--------|
| Logistic Regression | {'logreg_C': 10, 'logreg_penalty': 'l2'}                                         | 0.2225 |
| k-Nearest Neighbors | {'knn_n_neighbors': 3}                                                           | 0.2759 |
| Decision Tree       | {'tree_max_depth': 4, 'tree_min_samples_leaf': 25, 'tree_min_samples_split': 50} | 0.2706 |

Table 1. Cross-validated F1 scores and best hyperparameters for all models.

### 13.4. Test Set Comparison

| Model               | Test Accuracy | Test F1 | Test Recall | Test ROC-AUC | Best k |
|---------------------|---------------|---------|-------------|--------------|--------|
| Logistic Regression | 0.8513        | 0.2319  | 0.1554      | 0.8019       | —      |
| k-Nearest Neighbors | 0.8191        | 0.2912  | 0.2429      | 0.6661       | 3      |
| Decision Tree       | 0.8506        | 0.2903  | 0.1999      | 0.7817       | —      |

Table 2. Test set performance metrics for all models.

### 13.5. Visualizations

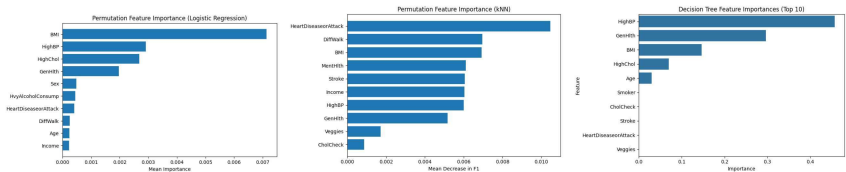


Figure 1. Permutation Feature Importances: Logistic Regression, kNN, Decision Tree (Top 10)

## Milestone 1 - Data Description

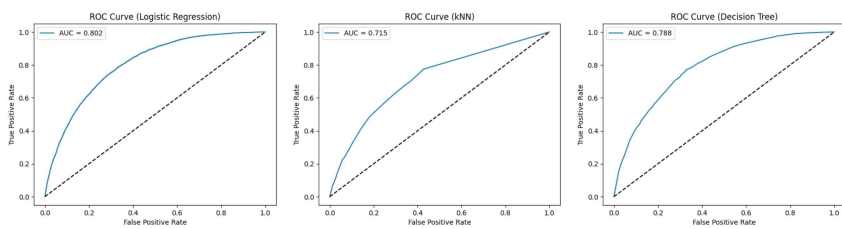


Figure 2. ROC Curves: Logistic Regression, kNN, Decision Tree

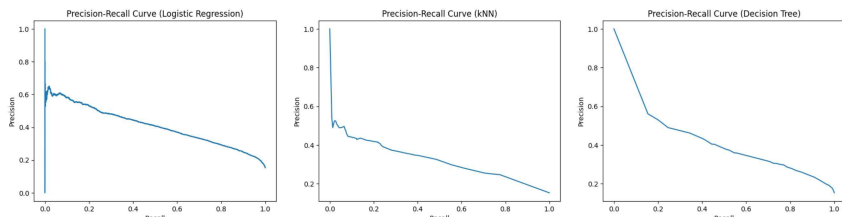


Figure 3. Precision-Recall Curves: Logistic Regression, kNN, Decision Tree

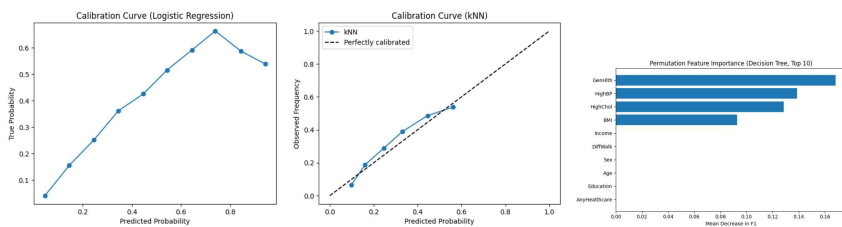


Figure 4. Calibration Curves: Logistic Regression, kNN, Decision Tree

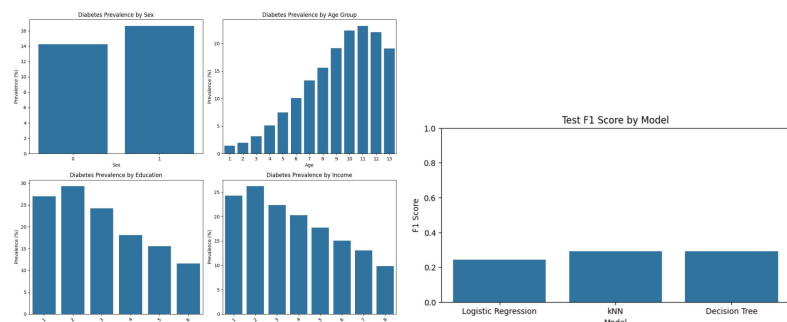


Figure 5. Demographics and Test Metrics: Diabetes Prevalence by Group, Test ROC-AUC by Model

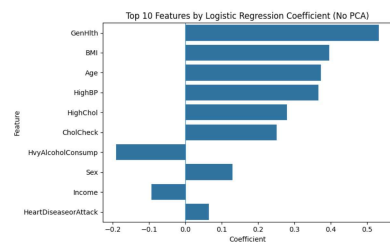


Figure 6. Top 10 Features by Logistic Regression Coefficients (No PCA)

### 13.6. Summary of Findings

- **k-Nearest Neighbors (kNN)** achieved the highest Test F1 (0.2912), making it the best model by our selection

rule. Accuracy is slightly lower than logistic regression (0.8191), and ROC-AUC is moderate (0.6661). Optimal  $k = 3$  neighbors. Permutation importance highlights the same top variables as logistic regression: general health, BMI, age, high blood pressure, and heart disease history. kNN is computationally expensive and less interpretable.

- **Logistic Regression** has high accuracy (0.8513) due to class imbalance; F1 (0.2319) and recall (0.1554) are moderate. ROC-AUC is high (0.8019), indicating meaningful signal. Calibration curve is reliable; key predictors align with clinical expectations.
- **Decision Tree** shows slightly lower F1 (0.2903) and ROC-AUC (0.7817). Accuracy is high (0.8506) due to class imbalance. Limiting depth and leaf size reduces overfitting but lowers flexibility. Provides interpretable rules identifying the same core risk factors.
- **Comparison Observations:** Accuracy is not very informative due to class imbalance; F1 and ROC-AUC are more meaningful. kNN has the highest F1, logistic regression has the best calibration and ROC-AUC, and the decision tree provides interpretable rules. Cross-validation F1 scores are slightly lower but show similar relative performance ranking.
- **Demographics:** Diabetes prevalence rises with age, is higher in lower education and income groups, and shows small sex differences. These patterns match the top features identified by all models.

## 14. Initial Result Analysis

Among the three models, k-Nearest Neighbors (kNN) achieved the highest F1 score, indicating the best balance between precision and recall. Its performance on the test set makes it the top model according to our selection criteria, with an optimal configuration of  $k = 3$  neighbors. Although its accuracy is similar to logistic regression, its ROC-AUC is moderate (around 0.66), suggesting that while ranking quality is reasonable, it does not significantly outperform logistic regression in discriminative ability. Permutation feature importance for kNN highlights the same key predictors as logistic regression, including general health, BMI, age, high blood pressure, and prior heart disease. However, kNN is computationally more intensive and less interpretable, making it better suited as a benchmark model rather than for direct deployment.

Logistic regression exhibits stable but moderate performance. Its high accuracy (0.85) is largely influenced by class imbalance rather than superior predictive capability. F1 and recall scores are modest, indicating the model still misses many positive diabetes cases. The ROC-AUC of

0.80 demonstrates meaningful signal detection, and the calibration curve suggests predicted probabilities are reliable. Key drivers identified by the model align with clinical expectations.

The decision tree shows slightly lower F1 (0.29) and ROC-AUC (0.78) compared to the other models. While its accuracy appears high, this is primarily due to class imbalance. Constraining tree depth and minimum leaf size reduces overfitting but also limits flexibility. Nevertheless, the decision tree consistently identifies the same core risk factors as the other models. Its main advantage is interpretability, providing simple, human-readable rules rather than top predictive performance.

Comparing all models, accuracy alone is not a meaningful measure due to class imbalance. F1 and ROC-AUC offer a more informative assessment of performance. kNN leads on F1, logistic regression performs best on ROC-AUC and calibration, and the decision tree is most interpretable. Cross-validation F1 scores are slightly lower than test set F1s, indicating consistent model behavior across folds, with logistic regression and kNN performing closely and the decision tree lagging.

Demographic analysis reveals that diabetes prevalence increases sharply with age, is higher in lower education and income groups, and shows smaller differences by sex. These patterns are reflected in the predictors deemed most important by all three models.

Overall, the three modeling approaches consistently identify the same key predictors: general health, BMI, age, blood pressure, and prior heart disease, confirming the main signals and trends in the dataset.